The Last Mile Is an Idea: Cognitive Operators, Proposer Strength and the Limits of Office-Scale Verified Search

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Abstract

Verified search — a language model proposing programs, a hard evaluator scoring them, selection keeping the best — is how machine systems have recently moved mathematical records. The published systems vary the scaffolding (islands, model ensembles, novelty machinery) and report the wins; none measures, with controls, which parts of the *proposer's cognition* matter. We instrument a minimal FunSearch-style loop at office scale (a 30B local model on a laptop, 120–600 verified samples per run) with three operators isolated in our earlier work on open-ended generation: a *schematic notebook* the model writes and carries instead of verbatim elite programs, a *named obstacle* (agenda), and *behavioural repulsion* from constructions already found. On nine construction problems from a public repository of open problems (six in the original battery, three added in a pre-registered extension), an ablation of five arms — each operator alone and their triple composition, two replicates — completed to the full 2^3 factorial certifies the primary contrast: the composed operators close a larger fraction of the seed-to-record gap (+0.196 pooled $[+0.059, +0.343]$, $p = 0.023$, exact paired sign-flip over nine problems, two replicates); behavioural repulsion raises functional diversity in all nine problems and both replicates (+0.31, $p = 0.0039$), with the caveat that the operator optimizes a quantity related to this outcome; and the completed 2^3 rejects anchoring as an interaction (memory×repulsion $\beta \approx 0$, $p = 0.47$): the certified composite gain decomposes additively, with schematic memory the largest single share (∼58%; raw $p = 0.047$, not surviving correction at $n = 9$), and anchoring survives only in the tail — memory+repulsion never collapses in eighteen runs, against three for repulsion alone and four for agenda+repulsion. At a 5× budget the composition keeps climbing while repulsion alone plateaus. Swapping the proposer for a frontier model (Claude Opus 5) under the identical loop reaches in tens of samples what the local model does not reach in hundreds — locating the residual gap in the proposer's construction knowledge, not in the loop; tripling the verifier's time budget then moves the result by +0.0002. Across proposers, grids and time budgets the search stalls at ≈92% of the best published lower-bound construction [1], and the candidate programs of both proposers remain within a sparse-atom construction family. Office-scale verified search reaches 57–99.9% of the reference values on these problems (median 89%); we state as a hypothesis, and design a test for, the reading that the remaining gap is a matter of construction family — an idea — rather than budget. We release the harness, every candidate, and the dated pre-registrations.

1 Introduction

FunSearch [11] and AlphaEvolve [8] demonstrated that a language model inside an evolutionary loop with a hard evaluator can produce genuinely new mathematical constructions. The systems that followed improved the *scaffolding*: sample-efficient archives and novelty rejection [5], verbal reflections as search gradients [13], stronger databases and routing. Two questions have stayed unmeasured. First, *which parts of what the proposer is asked to think* — its memory of the lineage, its statement of the obstacle, its relation to what was already found — change the search, and by how much? The prompts of the published systems show elite programs verbatim and vary everything else. Second, *how much of the outcome is the proposer itself*, holding the loop fixed? Published systems change model and scaffold together.

This paper measures both, at a scale any laboratory can afford, with the discipline our earlier studies [9] found necessary: every battery pre-registered before running, dated amendments, exact paired tests, and negative results reported at the same volume as positive ones. The operators we test are not arbitrary: they are the three interventions that survived a two-paper program on open-ended generation — a *schematic memory* (the model's own compressed recap outperformed verbatim context on document-level integration), a *standing question*, and *repulsion* from the already-produced, which in weight space collapsed without an anchor and in behaviour space is the natural tail-preserving form.

Contributions. (1) A pre-registered program of cognitive operators inside a fixed verified-search loop, grown from a 5×6×2 ablation to the complete $2^3 \times 9 \times 2$ factorial: the composition certifies at +0.196 ($p = 0.023$), behavioural repulsion buys functional diversity in every problem and replicate ($p = 0.0039$), and no single operator is safe alone. (2) A revision, by the completed factorial, of this program's recurring *anchoring pattern*: the memory×repulsion interaction on mean gap closed is zero — anchoring is not superadditive gain but *collapse prevention*. Repulsion without memory still breaks (three collapses in eighteen runs; four when paired with agenda), while memory+repulsion never does, echoing unanchored preference optimization in weight space. (3) A *scaling inversion*: from a 120-sample tie, the composed arm climbs to 0.3816 at 600 samples while repulsion alone stalls at 0.3514 — diversity without accumulated direction does not compound. (4) A controlled *proposer comparison*: local 30B versus a frontier model under the identical loop and budget, on the same problem — locating the residual gap in the proposer rather than in the loop, with a frontier-strength scoping run in which the operators' mean contribution sits inside run-to-run noise. (5) An honest calibration of office-scale verified search — 57–99.9% of the reference values (median 89%) at 120–600 samples — and a characterized wall: grid resolution, verifier time and proposer strength each ruled out, leaving the construction family itself as the binding constraint.

2 Related work

Verified search systems. FunSearch evolved single functions with small code models and millions of samples [11]; AlphaEvolve evolves whole files with frontier ensembles and thousands [8], and its repository of 67 problems with official verifiers [3] is the substrate of our experiments. ShinkaEvolve reports record-level results with ∼150 samples using novelty-based rejection and bandit model selection [5]; ReEvo uses model-written reflections as "verbal gradients" [13]. Our schematic notebook differs from ReEvo's reflections in being a persistent, compressed lineage memory that *replaces* verbatim elites rather than commenting on pairs; and none of these systems reports a controlled ablation of its own prompt components. LoongFlow [12] composes planning, hybrid memory and MAP-Elites niches into one system and reports large efficiency gains; it is the closest architecture to our composed arm, and exactly the kind of system whose components an ablation like ours aims to isolate.

Why diversity needs help. Without selection pressure, LLM mutation chains collapse into attractor regions [4]; RL-tuned models lose pass@k coverage relative to their base [14]; and coverage under repeated sampling grows log-linearly only when a verifier exists [2]. Our behavioural-repulsion operator is novelty search [6, 7] transplanted into the proposer's prompt, with the construction itself (not an embedding) as the behaviour descriptor.

3 The harness, the operators, the problems

Loop. A minimal FunSearch-style loop: K islands, each holding its best programs; per generation, per island, a prompt is built and S candidates sampled; each candidate runs in a sandbox (time and memory limits) and is scored by the problem's verifier; islands keep bounded elites and migrate their best every third generation. The proposer is `Qwen3-Coder-30B-A3B-Instruct` (8-bit, MLX, on a MacBook M5 Pro) prompted through its chat template; Section 5 swaps in Claude Opus 5 through an API under the identical loop. Every candidate — code, score, error, generation, island, and a behaviour signature (a hash of the rounded construction it outputs) — is logged.

Operators. **A** (baseline): the prompt shows the statement and the island's top-3 programs with scores, FunSearch-style. **B** (schematic notebook): each generation the model writes a $\leq$12-line notebook — what worked, what failed and why, the open question — from the elites and the recent failures; the prompt then shows the notebook and only the single best program. **C** (agenda): one model-written line naming the structural obstacle of the current best, appended to the prompt. **D** (behavioural repulsion): candidates whose construction duplicates an island member's are rejected, and the prompt lists already-found constructions as forbidden. **E**: B+C+D.

Problems. Nine construction problems ported from the official repository notebooks, chosen for cheap exact verification and known headroom. The original battery: circle packing (26 circles, maximize the sum of radii), ring loading (m=15, exact 2^m verification), beat-the-average (a pmf maximizing $P[X_1{+}X_2{+}X_3 < 2X_4]$), the first autocorrelation inequality (minimize an upper bound on C_1), isosceles-free subsets of the 64^2 grid, and sum-difference III. A pre-registered extension, run before any n=9 analysis, added three: the max–min pairwise-distance ratio for 16 points in the plane (minimize), the Heilbronn triangle problem for 11 points (maximize the smallest triple area), and splitting 180! into 180 factors maximizing the smallest (exact big-integer verification). Each run's budget is 120 verified samples (10 generations $\times$ 2 islands $\times$ 6); the primary metric *frac* is the fraction of the seed-to-record gap closed, *auc* its mean over the budget (sample efficiency), *div* the share of distinct behaviours among valid candidates.

4 The factorial

Pre-registered contrasts (exact paired sign-flip over problems, cells = problems, pooled replicates). The original six-problem battery put the primary contrast at the design's resolution floor (+0.199, $p = 0.0625$; with six cells and one adverse problem the exact test cannot go below 4/64); the pre-registered nine-problem extension certifies it: **F1** (primary), E vs A on frac: +0.196 [+0.059, +0.343], $p = 0.023$. **F4**, D vs A on functional diversity: +0.307 [+0.149, +0.477], $p = 0.0039$ — the minimum attainable p, positive in all nine problems in both replicates. **F6** (the anchor), E vs D: +0.137 [-0.047, +0.382], $p = 0.156$ — not supported as a pairwise contrast; the factorial below resolves what it was measuring. F2 (auc): +0.115, $p = 0.070$: the operators cost budget early (the notebook and agenda calls spend samples' worth of tokens) and repay it late.

frac (pooled over 2 replicates)	A	B	C	D	BC	BD	CD	E
circle packing 26	0.703	0.974	0.976	0.000	0.989	0.752	0.980	**0.992**
ring loading 15	0.444	0.767	**0.808**	**0.808**	0.731	**0.808**	**0.808**	**0.808**
beat-the-average	0.181	0.384	0.519	0.608	0.661	**0.789**	0.540	0.709
autocorrelation C_1	0.059	0.156	0.041	0.202	0.014	**0.212**	0.000	0.104
isosceles-free 64^2	**0.560**	**0.560**	**0.560**	0.399	**0.560**	**0.560**	**0.560**	**0.560**
sum-difference III	**0.392**	0.329	0.253	0.158	0.314	0.327	0.069	0.362
max–min distance 16	0.169	0.992	0.372	**0.993**	0.919	0.793	0.247	0.708
Heilbronn triangle 11	0.672	0.431	0.507	0.540	0.371	0.477	0.619	**0.700**
180! into 180 factors	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**
mean frac	0.439	0.596	0.534	0.498	0.592	0.610	0.511	**0.635**
mean functional diversity	0.41	0.53	0.52	0.72	0.59	0.68	**0.80**	0.70
collapses (frac < 0.05, of 18 runs)	2	1	1	3	2	**0**	4	1

Table 1: The complete 2^3 factorial over nine problems, pooled over two independent replicates (120 verified samples per run). frac = fraction of the seed-to-record gap closed. E has the best mean and is best or tied-best in five of nine problems; D alone is bimodal — best on wide valid manifolds (pmfs, sequences), catastrophic where validity is fragile (packing geometry, constrained integer sets); BD is the only arm that never collapses, CD the one that collapses most; two problems (isosceles-free, 180!) are family plateaus where every arm meets the same wall.

The completed 2^3. Three pairwise arms (memory+agenda, memory+repulsion, agenda+repulsion) over the same nine problems and two replicates complete the cube (8 cells $\times$ 9 $\times$ 2; pre-registered before running). The effect-coded analysis (exact sign-flip over problems, 2^9) rejects the primary interaction: memory$\times$repulsion $\beta = +0.003$ [-0.039, +0.035], $p = 0.47$ one-sided; agenda$\times$repulsion is null (-0.007, $p = 0.79$); and no main effect survives BH correction — schematic memory is the largest single term ($\beta = +0.057$ [+0.014, +0.105], raw $p = 0.047$, $q = 0.14$), agenda (+0.016) and repulsion (+0.011) near zero on frac. The decomposition is exact ($E - A = 2\,(\beta_M + \beta_A + \beta_R + \beta_{MAR}) = 0.196$): the certified composite gain splits as roughly 58% memory, 16% agenda, 12% repulsion and 14% three-way — at $n = 9$ the total certifies, the parts do not. Two patterns in the cube are worth the trip. First, each operator has one job (Table 1, Figure 1): the four memory-on cells hold the four best mean fracs, while every repulsion-on cell sits at 0.68–0.80 functional diversity against 0.41–0.59 without it — memory moves the gap, repulsion moves diversity. Second, the anchor lives in the tail, not the mean: memory+repulsion is the only arm that never collapses (0 of 18 runs, against 3 for repulsion alone and 4 for agenda+repulsion, the cube's worst), so what memory buys repulsion is insurance against starving the island, not superadditive mean gain. Agenda, decomposed at last, pays for neither: near-zero on frac alone, and the worst collapse count when paired with repulsion.

The heterogeneity is the finding the factorial adds to the literature: *where* the valid manifold is wide (a pmf, a sequence of pairs), repulsion alone is the best single operator; where validity is fragile (non-overlapping circles, exact integer constraints), it starves the island (both replicates of circle packing collapse) — and the composed arm inherits most of the benefit with none of the catastrophes. No single operator is safe; the composition is.

5 Scale, proposers, and the wall

We then took the problem with guaranteed headroom — beat-the-average, where AlphaEvolve's own reported value (0.3890) sits below the best known (0.400695) — and ran a pre-registered ladder. (i) *Budget*: at 600 samples the composed arm reaches 0.3816 while repulsion alone, which had tied it at 120 samples, stalls at 0.3514: diversity without accumulated direction does

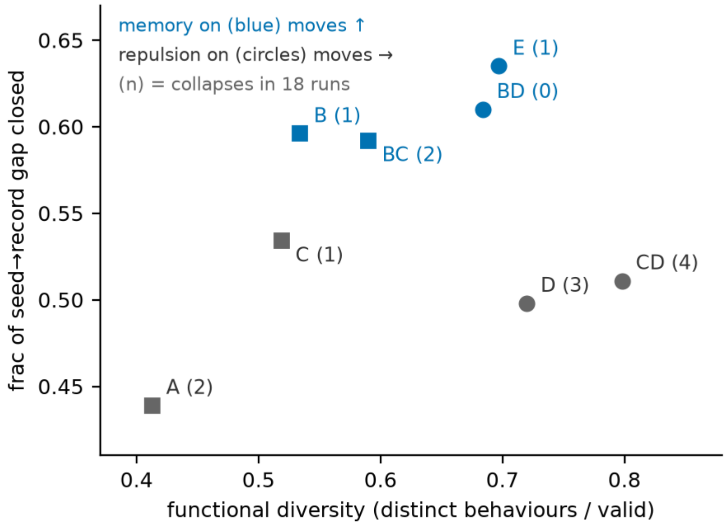

Figure 1: The completed 2^3 at a glance (cell means over nine problems, two replicates). Memory-on cells (blue) hold the four best gap fractions; repulsion-on cells (circles) hold the highest functional diversity; the two axes barely interact ($\beta_{M\times R} \approx 0$). Collapse counts in parentheses: memory+repulsion is the only arm that never collapses; agenda+repulsion collapses most.

not compound. (The same 5× extension on max–min distance 16 left both local arms *worse* than their own 120-sample runs on a fresh seed — seed variance dominates budget there too.) (ii) *Grid*: refining the pmf grid from L=5000 to L=20000 at equal budget makes the local proposer *worse* (0.3561): the limit was not resolution. (iii) *Proposer*: Claude Opus 5 under the identical loop reaches 0.3897 within ∼100 samples — just above the repository's published value, which was obtained with 2025-era proposers; we read this as locating the wall, not as ranking systems across model generations — and plateaus there for 17 generations. (iv) *Verifier time*: tripling the sandbox budget to 60 seconds, so heavier optimizers can run, moves the plateau by +0.0002.

The constructions are legible. The local model's best is a bimodal sparse measure of 16 atoms — half the mass on $\{0, 1, 2, 3\}$, geometrically spaced atoms at the top — with the strategy stated in comments. Claude's best implements an exact sparse-atom objective with an analytic gradient and a multiplicative-weights refinement. Both proposers, and AlphaEvolve's published value, stall within half a percent of each other, at ≈92% of a record whose construction evidently lies outside the sparse-atom family every language model proposes. Grid, time and proposer strength are ruled out; what remains is the family — an idea.

A second problem sharpens where such ideas live. On max–min distance 16, the frontier proposer under the same loop emits, *in generation zero* — before any search feedback — a minimax program (SLSQP on $\min t$ s.t. $d_{ij}^2 \le t$, $d_{ij}^2 \ge 1$, analytic Jacobians) whose value lies marginally below the repository's published one; every valid sample across the run converges to the same value. The loop verified this construction; it did not discover it. We treat the datum as scope, not as a record claim: the operators structure search and protect its tails, but they do not inject construction knowledge the proposer lacks. A pre-registered scoping run makes the same point from the other side, on beat-the-average: at frontier strength the bare, memory-only and repulsion-only arms land within 0.001 of one another (0.3871, 0.3866, 0.3876; single runs, estimation only) and within 0.003 of the composed arm's 0.3897 — the operators' certified gains are a local-proposer phenomenon, and their mean contribution at frontier strength on this problem sits inside run-to-run noise (whether the collapse insurance survives at this strength remains untested; the two remaining scoping problems were cancelled for focus before any of these results were seen, as dated in the pre-registration).

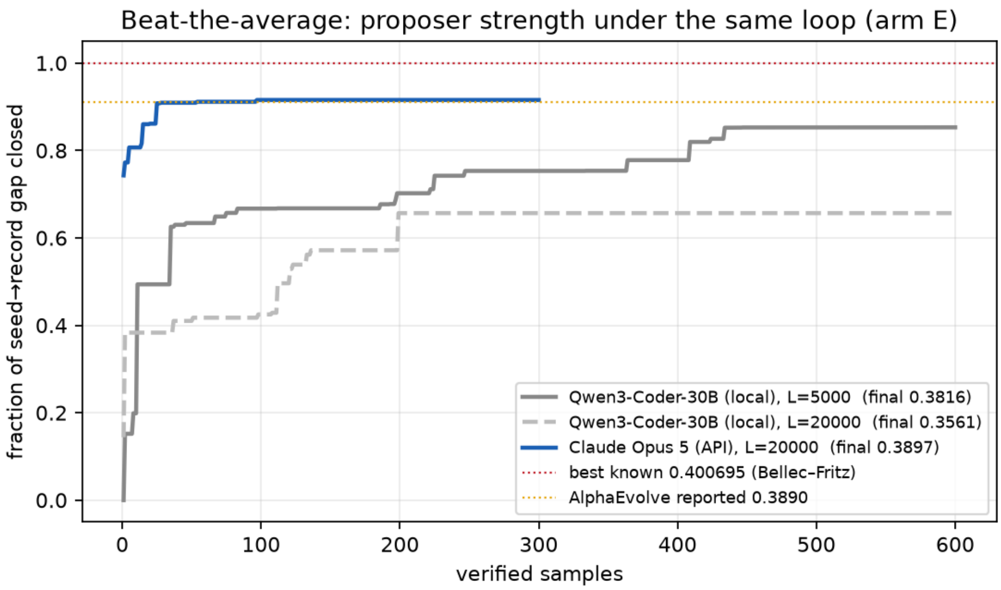

Figure 2: Beat-the-average under the same loop (arm E): fraction of the seed-to-record gap closed per verified sample. The local 30B at two grid resolutions; Claude Opus 5; dotted lines mark AlphaEvolve's reported value (0.3890) and the best known (0.400695, Bellec–Fritz).

6 Discussion

What office scale buys. With 120–600 verified samples, a local model and one laptop: 57–99.9% of the best known values (median 89%) across nine problems, legible programs, and — with a frontier proposer at ∼US$50 of API — parity with the values the large systems report on this problem. The published record-movers used frontier ensembles with thousands of samples [8], millions with small models [11], or open-source loops with strong API models [5]; the band we measure — local proposer, hundreds of samples, controlled operators — was unoccupied, and it is the band most laboratories can afford.

The operator ledger. Composition is the only arm that never collapsed, and its edge has a mechanism visible in the notebooks: the schematic memory keeps direction across generations, the agenda names the obstacle, and the repulsion spends the saved budget on genuinely different constructions. Alone, each fails somewhere: memory and agenda are inert on problems the model already saturates; repulsion starves fragile manifolds. This is the fifth time this program has measured the same law — pushing away from something only works while anchored to something — across narrative generation, weight-space preference optimization, and now prompt-space search.

The proposer comparison and the plateau. On this one problem, the frontier proposer is far more sample-efficient (+0.008 absolute over the local model's 600-sample best, reached 20× faster), and everything we tried after it moved little (grid, time, more samples: +0.0002). This is a sample-efficiency comparison on one problem under one harness — not yet a proposer-strength curve; replications on further problems are planned, and reasoning effort, ensembles, alternative representations and thousand-sample budgets all remain untested explanations. The shared plateau suggests, consistently with our consolidation results [10], that search over a fixed prior rapidly collects what the prior's construction families contain and then stalls at the best family member — a hypothesis to test rather than assert. *The family-hint experiment* (registered as the next battery), on beat-the-average: (a) no hint; (b) a generic hint ("try a different representation"); (c) a structural hint naming the family of the Bellec and Fritz [1] construction without parameters; (d) a seed from that family far from its optimum; (e) the full construction as an oracle control. If the loop optimizes the family once given but never proposes it unaided, the missing ingredient is the idea; if it cannot even optimize it, the story is different. This is the question Georgiev et al. [3] report answering with an expert in the loop, from the other side.

Design rules. For verified search at small scale: compose the operators, never run repulsion alone on fragile manifolds; keep the behaviour descriptor exact (the construction, not an embedding); spend on the proposer before spending on samples; treat plateaus as candidate family boundaries and test them — grid, time, proposer and hints in pre-registered steps — rather than declaring them.

7 Limitations

Nine problems is still a small universe of construction families; the full 2^3 isolates interactions, but at n=9 no per-term effect survives correction, so the operator decomposition (memory largest) is directional, not certified, and the collapse asymmetry is descriptive; budgets are equalized in verified samples, not tokens or calls (arms B, C and E spend one to two extra short model calls per island-generation; token accounting is logged from the factorial-completion runs); the diversity metric is related to the repulsion operator's own filter, and the behaviour signature's sensitivity to rounding is untested; the six problems satisfy criteria fixed before any runs (cheap exact verifier, short-function candidates, known headroom; the universe of 68 candidates and per-problem ratings are in the released reconnaissance table) but were not sampled randomly; *frac* is floored at the seed by construction, signs are harmonized for minimization, and reference values are frozen at the repository commit recorded in the release; the scaling ladder is one domain; the API proposer is not bitwise reproducible and its reasoning effort was fixed low; comparisons against repository values compare a 2026 proposer with values obtained by 2025-era systems, so they locate our wall and never rank systems — no system-level head-to-head with ShinkaEvolve or AlphaEvolve is attempted, and the operators should be read as structuring search, not as a substitute for construction knowledge the proposer lacks.

8 Conclusion

Inside a fixed verified-search loop at office scale, the cognitive operators our generation studies isolated — schematic memory, a named obstacle, behavioural repulsion — compose into the arm that was safest in these runs (one collapse in eighteen, against three for repulsion alone — and zero for memory+repulsion) and the one that kept improving with budget; repulsion alone enforces behavioural non-duplication by construction and was the best single operator exactly where the valid manifold is wide; and on the one problem where we compared them, a frontier proposer was more sample-efficient by an order of magnitude. Within every setting we tested, the search stalled at the boundary of the construction families the models propose. Whether crossing that boundary requires what we have informally called an idea is now a registered experiment rather than a metaphor.

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